Phase-Dependent Moisture–Convection Coupling in the Madden-Julian Oscillation

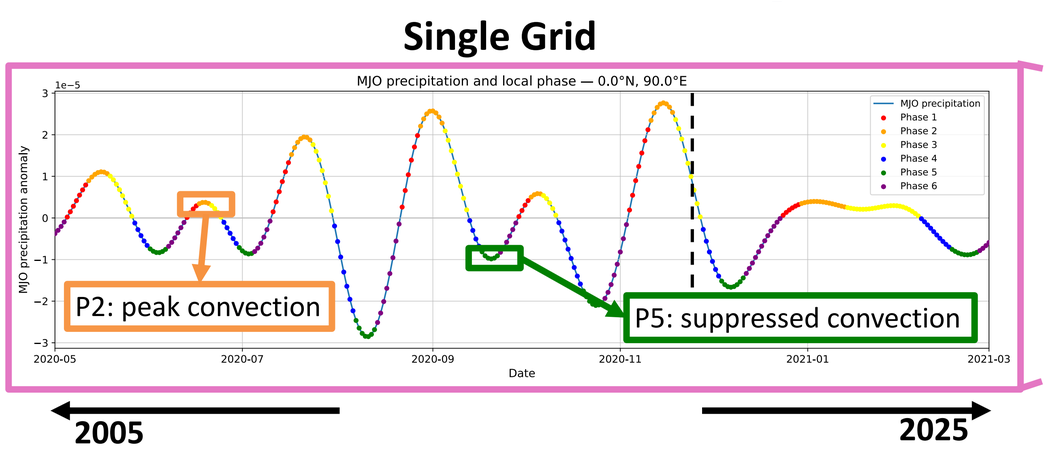
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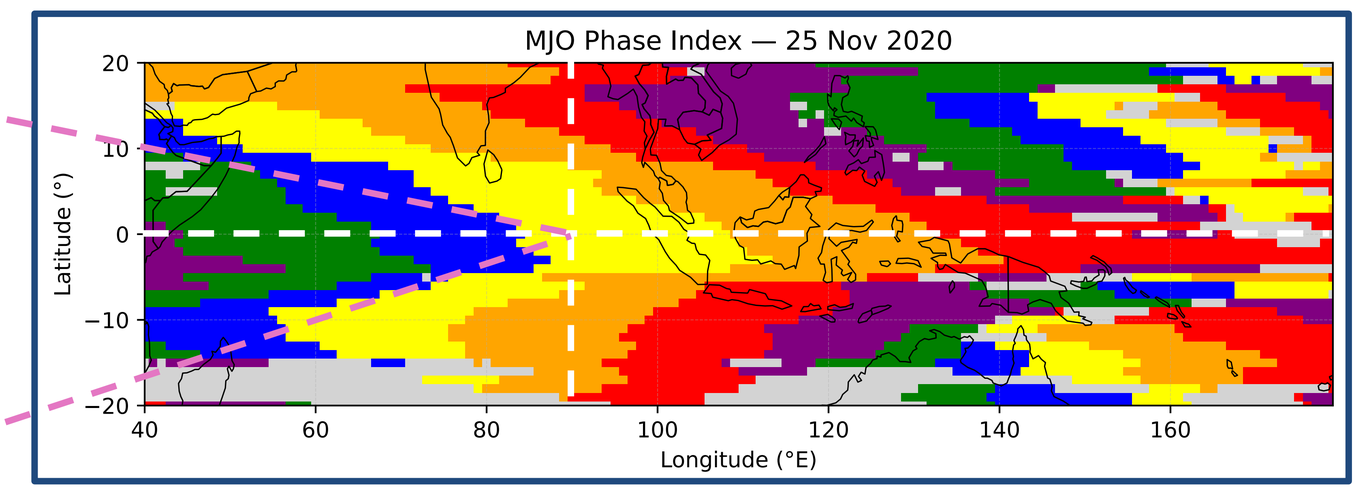
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Motivation and Goals

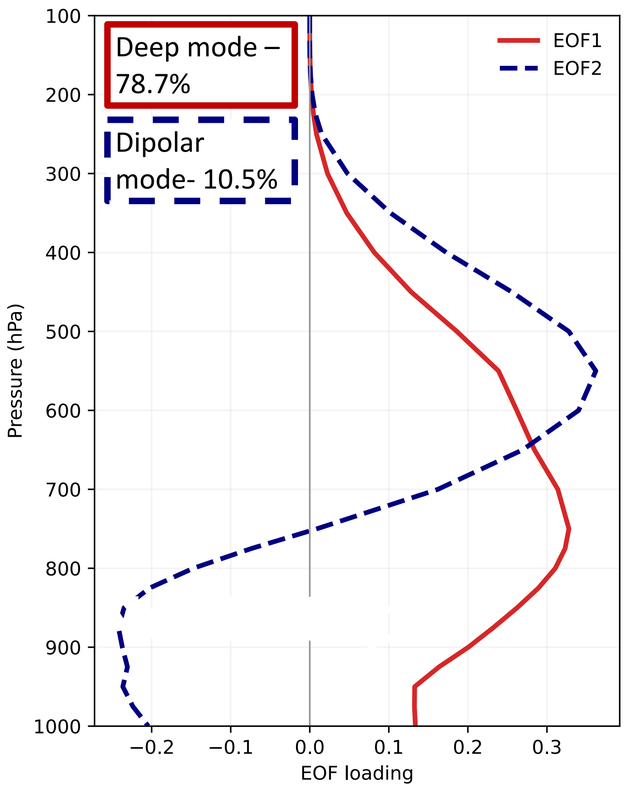
- Many numerical models struggle to reproduce the Madden–Julian Oscillation (MJO), partly because small-scale processes, particularly convection, are imperfectly parameterized. Understanding how these processes interact with MJO-scale moisture transport can help inform model development.
- Using ERA5 reanalysis, we combine a local precipitation-based MJO phase index with phase-resolved moisture budgets and scale decomposition to identify the horizontal and vertical transport —and interactions among climatological, MJO, and eddy components—shape MJO moisture evolution.

The MJO moisture life cycle





EOFs of MJO-scale q

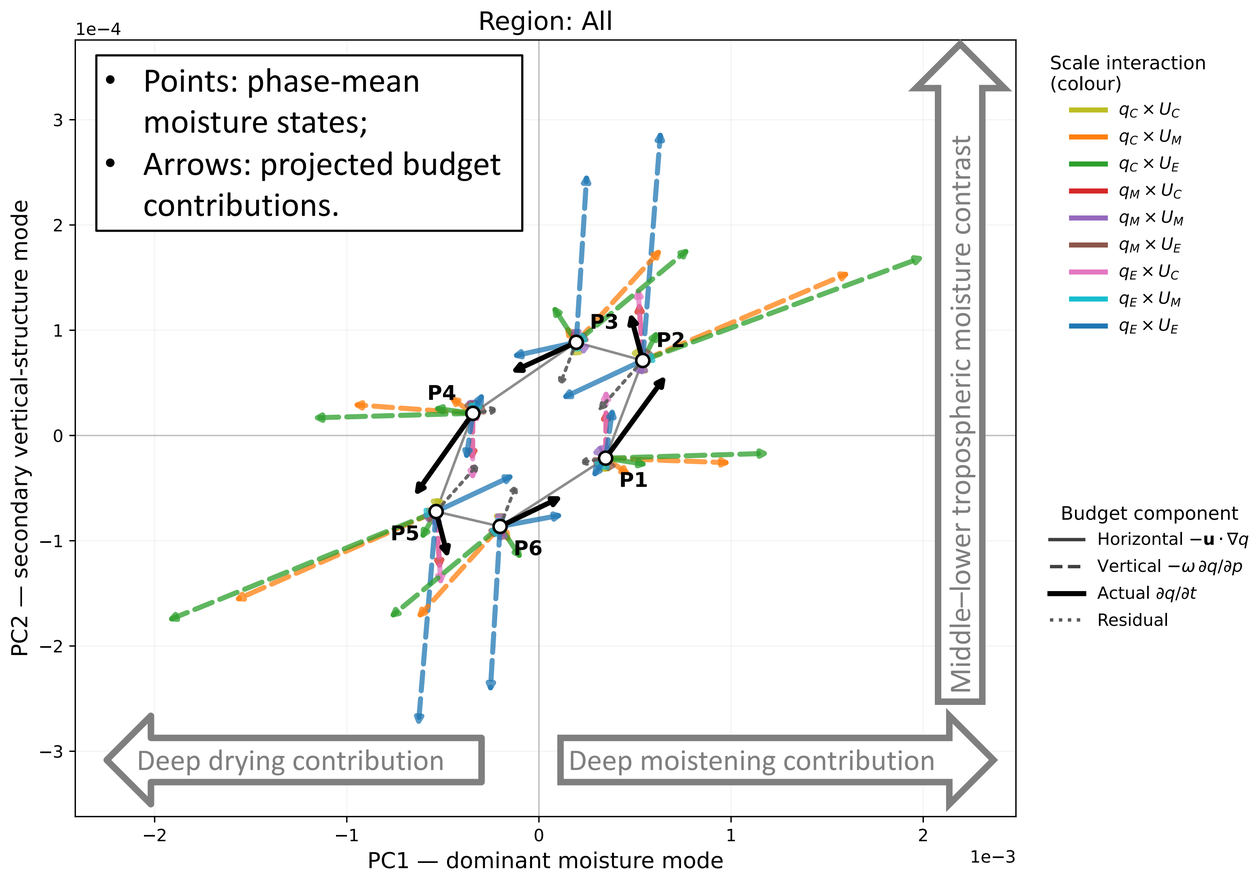


EOFs of MJO specific-humidity (q) anomalies; percentages show the variance explained. All phases and regions share the same basis.

2. Scale interactions underlying MJO moisture transport

$$\begin{cases} q = \bar{q} + q_{MJO} + q' \\ U = \bar{U} + U_{MJO} + U' \end{cases}$$

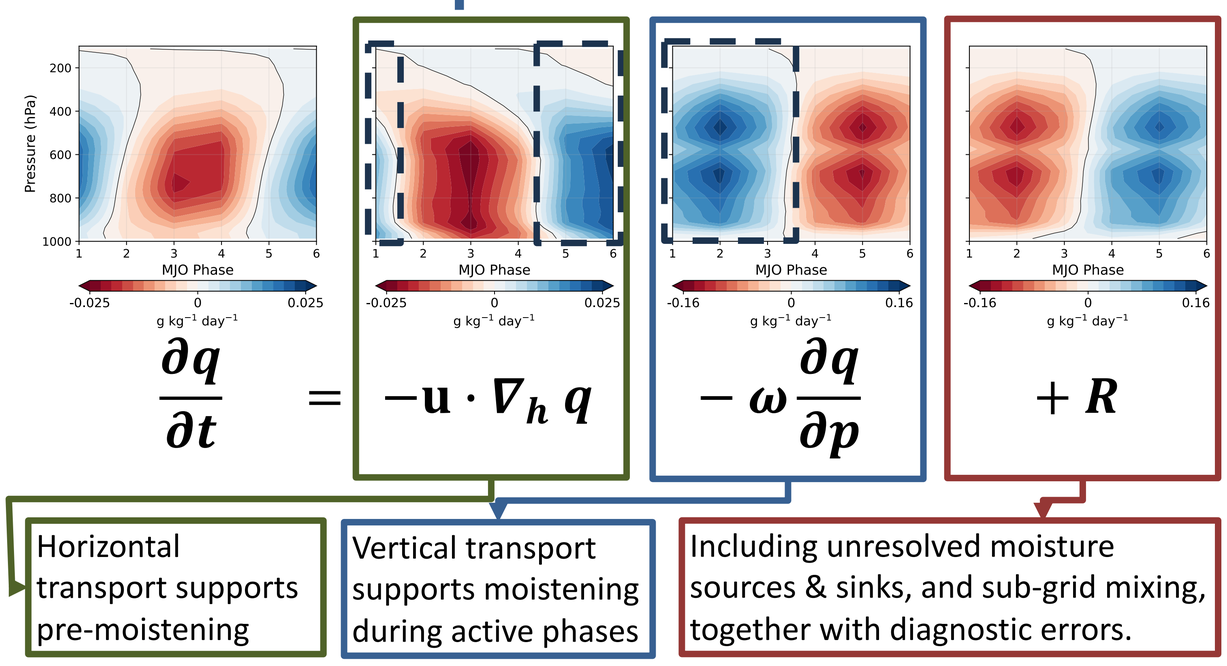
- Vertical transport of $q_C \times U_M$ and $q_C \times U_E$ provides prominent contributions along EOF1.
- $q_E \times U_E$ transport contributes strongly along EOF2.
- These contributions broadly reverse direction between P1–3 and P4–6, reflecting the phase dependence of moisture transport.



Note: Horizontal, vertical and dq/dt arrows are displayed at 10× the baseline arrow scale; residual arrows retain the baseline scale. Arrow lengths are not directly comparable between the residual and other terms.

C/M/E denote climatological, MJO, and eddy components. The Eddy is defined similarly to Tulich, S. N. (2023).

1. Moisture transport contributions across MJO phases



Conclusions.

- Horizontal transport supports pre-moistening; vertical transport supports active-phase moistening.
- Dominant scale interactions reverse with MJO phase, affecting moisture amplitude and vertical structure.

Next Steps.

- Extend the diagnostics to km-scale ICON simulation (res. Zoom 11), including explicit convective contributions.
- Use SCM experiments to test how well convection schemes reproduce MJO convective contributions.

1 Xu, C., Sherwood, S., Jucker, M. *et al.* (2026). The evaluation of convectively coupled waves in limited-length datasets. *Clim Dyn,* 64, 145. https://doi.org/10.1007/s00382-026-08106-2
2 Tulich, S. N. (2023). How important are horizontal eddy moisture transports for the MJO's eastward propagation? GRL, 50, e2023GL103944. https://doi.org/10.1029/2023GL103944